

ZOHO-2026-BIT

Round 2:

1. Minimum Operations to Make an Array Palindrome

Given an integer array, in one operation you can choose any contiguous subarray and add 1 to every element in it.

Find the minimum number of operations required to make the entire array a palindrome.

Sample Input

[1, 2, 3, 2, 1, 0]

Sample Output : 1

2. Spiral matrix

Given an $N \times M$ matrix and an integer K , rotate all the elements of the matrix in spiral order by K positions in the clockwise direction and print the resultant matrix.

Example

Input

4 4

1 2 3 4

5 6 7 8

9 10 11 12

13 14 15 16

3

Spiral Traversal

1 2 3 4 8 12 16 15 14 13 9 5 6 7 11 10

Rotate by $K = 3$

7 11 10 1 2 3 4 8 12 16 15 14 13 9 5 6

Output

7 11 10 1

6 13 9 2

5 14 15 3

16 12 8 4

3. Given an array of integers, sort the array into a wave like array and return it, arrange the elements into a sequence such that $a_1 \geq a_2 \leq a_3 \geq a_4 \dots$

Example:

Given [1, 2, 3, 4]

One possible answer : [2, 1, 4, 3]

Another possible answer: [4, 1, 3, 2]

4. Maximum Sum $K \times K$ Submatrix

Given an $N \times N$ matrix and an integer K , find the maximum sum of any $K \times K$ submatrix.

Input

$N = 3, K = 2$

1 2 3

4 5 6

7 8 9

Output

28

Explanation

The 2×2 submatrix:

5 6

8 9

$5 + 6 + 8 + 9 = 28$

5. Minimum Sum of Digits After Adding the Same Integer

Given an array A of N integers, define:

$f(x)$ = sum of the digits of x

You can choose any non-negative integer x and add it to every element of the array.

Find the minimum possible value of:

$f(A[0] + x) + f(A[1] + x) + \dots + f(A[N-1] + x)$

Sample Input 1

4

4 13 8 6

Sample Output 1

14 (when $x = 7$)

6. Given an array of integers, sort all the odd numbers in ascending order and all the even numbers in descending order, while preserving their original positions (odd numbers remain in odd positions and even numbers remain in even positions).

Input

5 2 8 1 7 4 3 6

Output

1 8 6 3 5 4 7 2

7. Given a string containing lowercase letters and digits, each digit represents the number of positions that the character immediately after it should move to the left.

Replace the digit with the moved character. If the character moves beyond the beginning of the string, use # as a placeholder.

Input

ab2x1c

Output

Abaxxc

8. Given an array of integers, sort the elements based on their frequency:

Elements with higher frequency come first.

If two or more elements have the same frequency, arrange them in ascending order.

Input

4 5 6 5 4 4 6 5 2

Output

4 4 4 5 5 5 6 6 2

9. Given a String, remove the palindromic words from the string and print the remaining string.

Input: I love malayalam language

Output:

I love language

10. Given N strings of the same length, compare the characters at each position.

If the same character appears in all strings at a position, print that character.

Otherwise, print the character that appears most frequently at that position.

If two or more characters have the same maximum frequency, print *.

Also print:

The number of positions where all strings have the same character.

The number of positions where one character has the maximum frequency.

Input

4

team

teal

teem

tear

Output

team

2

2

Another example:

Input:

4

team

teel

teak

tear

Output:

tea*

2

1